CSP project

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1 Introduction

The current global energy transition is increasingly driven by the urgent need to decarbonize the power sector. While variable renewable energy sources like wind and photovoltaics have seen massive growth, their inherent intermittency poses a significant challenge for grid stability and reliability. In this scenario, Concentrated Solar Power (CSP) emerges as a strategic technology. Unlike direct generation methods, CSP possesses the unique capability to decouple solar energy collection from electricity generation through the integration of cost-effective Thermal Energy Storage (TES). This feature allows CSP plants to be fully dispatchable, providing firm capacity and shifting energy production to cover peak demand periods even after sunset, thereby stabilizing the grid and reducing the reliance on fossil-fuel backup. [24]

Our project focuses on the design of a Solar Tower. The plant configuration consists of a heliostat field concentrating Direct Normal Irradiance (DNI) onto the receiver tower, where a heat transfer fluid (molten salts) is heated and stored in a two-tank direct system before feeding the power block.

The design methodology adopts a specific optimization approach based on a fixed solar field layout. Starting from a pre-defined field configuration we evaluated the optical performance of the heliostats and quantifying critical losses. The optimization focuses on the Power Block and TES sizing. The goal is to determine the optimal Solar Multiple and storage capacity that maximize the system's capacity factor and minimize the Levelized Cost of Electricity (LCOE), ensuring the most efficient utilization of the available solar resource.

2 Location

For the deployment of the solar tower plant, we selected a site located in Jean, Nevada,(Coordinates: 35°45'N 115°21'W) approximately 30 miles south of Las Vegas. This location represents a strategic choice driven by its exceptional topographical and meteorological characteristics.
The area is characterized by extensive flat terrain and vast, unobstructed spaces. These features are essential prerequisites for accommodating the large footprint of required by the solar field [4].

The suitability of this region for large-scale solar energy exploitation is further validated by the industrial context of the neighboring area of Primm. This zone already hosts other solar plants, including large-scale PV facilities, confirming both the abundance of the solar resource and the feasibility of land use for energy infrastructure.

3 Solar Field

Our solar field layout was optimized to maximize the energy collection while minimizing shading and blocking effects.The final configuration consists of 3615 heliostats, covering a total reflective area of 521915 m^2. As illustrated in Figure 1, SolarPilot simulations determined that the optimal field layout follows a radial staggered pattern symmetric along the East-West axis, characterized by a variable density that increases towards the tower to maximize land use efficiency.

Input data on SolarPilot was tower height:195m, solar field design power 350MW, design DNI $950 W/m^2$, sun location at design point at summer solstice. Performance simulations was carried out as summer solstice , solar noon in order to design the plant at highest irradiation possible. Other data included receiver height:15m, receiver diameter:17.65 m .

To accurately evaluate the plant's performance on a yearly basis, a Python-based model was developed to determine the sun's position for every hour of the year. The code iterates through 365 days (n) and 24 hours (h), calculating the key orbital parameters essential for the optical analysis. Specifically, the model computes the Solar

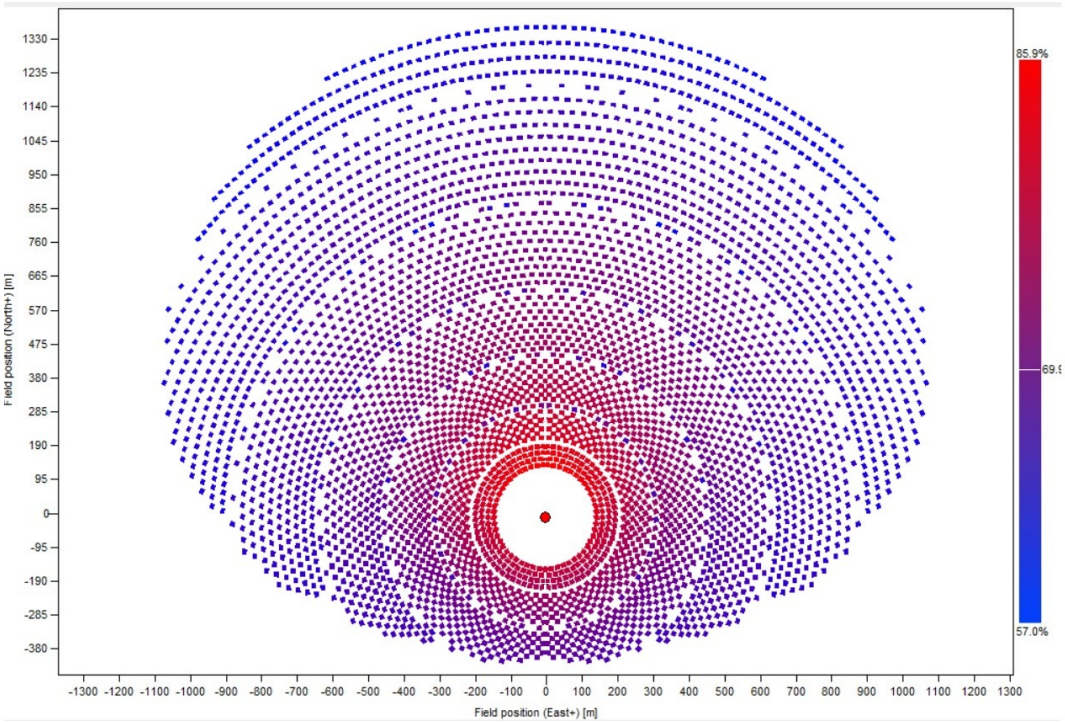


Figure 1: CSP layout

Declination (δ) and the Equation of Time (E_n) to account for the eccentricity of Earth's orbit and the obliquity of the ecliptic. By correcting the standard time with the longitude difference and the equation of time, the Local Solar Time (t_s) and the Hour Angle (ω) are derived.

1. Equation of Time

$$E_n = 229.18\Big[0.000075 + 0.001868\cos\left(\frac{360(n-1)}{365}\right) - 0.03277\sin\left(\frac{360(n-1)}{365}\right) \tag{1}$$

$$- 0.014615\cos\left(2\,\frac{360(n-1)}{365}\right) - 0.04080\sin\left(2\,\frac{360(n-1)}{365}\right)\Big] \tag{2}$$

2. Standard Time Zone Longitude

$$\Phi_{STZ} = STZ \cdot 15^\circ$$

3.Local Solar Time

$$t_s = t + \frac{\Phi_{loc} - \Phi_{STZ}}{15} + \frac{E_n}{60}$$

4. Solar Hour Angle

$$\omega = 15 \cdot (t_s - 12)$$

5. Solar Declination Angle

$$\delta = 23.45 \cdot \sin\left(\frac{360(284+n)}{365}\right)$$

Finally, the script outputs the two fundamental coordinates: the Solar Altitude (α_s), representing the elevation of the sun above the horizon, and the Solar Azimuth (γ_s), indicating the sun's cardinal direction.

6. Solar Altitude Angle

$$\alpha_S = \arcsin(\sin\delta\sin\phi + \cos\delta\cos\phi\cos\omega)$$

8. Solar Azimuth Angle (raw)

$$\gamma_{raw} = \mathrm{sgn}(\phi) \cdot \arccos\left(\frac{\sin\alpha_s\sin\phi - \sin\delta}{\cos\phi\cos\alpha_s}\right) \tag{3}$$

It is important to note that a logical correction was applied to the raw azimuth values to ensure full consistency with the coordinate system adopted by SolarPilot. This alignment is a critical prerequisite for the subsequent modeling steps: since the optical efficiency is calculated by interpolating data explicitly exported from SolarPilot, the azimuth definition in the script must strictly match the software's convention to minimize interpolation errors.

$$\gamma_s = \begin{cases} 180° - \gamma_{\text{raw}} & \text{se } t_s \leq 12 \\ 180° + \gamma_{\text{raw}} & \text{se } t_s > 12 \end{cases}$$

The accuracy of this model is visually represented in Figure 2 , which plots the solar altitude against the azimuth at 12:00 PM for every day of the year. The resulting confirms the correct implementation of the orbital mechanics, highlighting the seasonal shift between the summer and winter solstices.

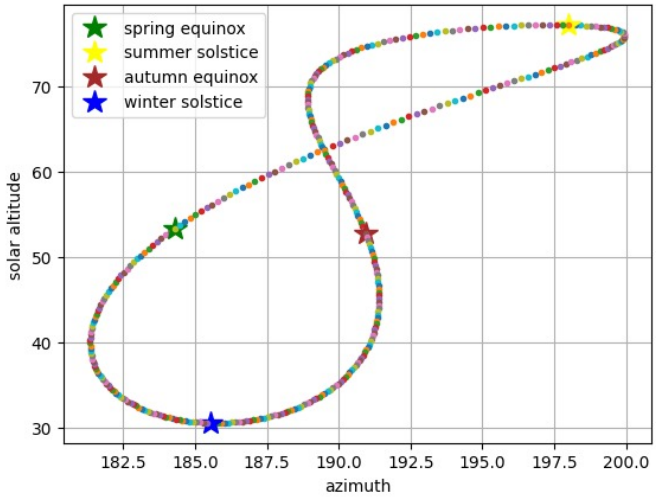


Figure 2: Altitude and azimuth at 12 AM throughout the year

With the solar coordinates (α_s and γ_s) accurately defined and aligned with the SolarPilot reference system, the next step involved the characterization of the solar field's performance. The optical efficiency (η_{opt}) is not constant; it varies significantly throughout the day and year due to geometric factors such as the cosine effect, shadowing, and blocking, as well as atmospheric attenuation.
To avoid the computational burden of running continuous ray-tracing simulations for every time step of the yearly analysis, a polynomial regression approach was adopted. Starting from a dataset of efficiency values simulated at various angles in SolarPilot, our Python script fit these data points into polynomial functions. Consequently, the total optical efficiency is expressed as a function of the solar position:

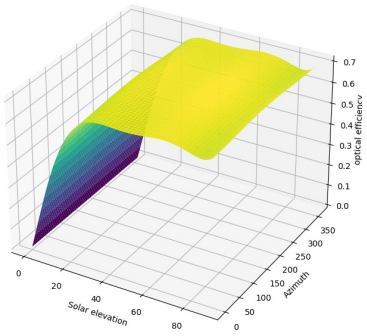
$$\eta_{opt}(\alpha_s, \gamma_s) = \eta_{des} \cdot f_{el}(\alpha_s) \cdot f_{az}(\gamma_s)$$

Where:

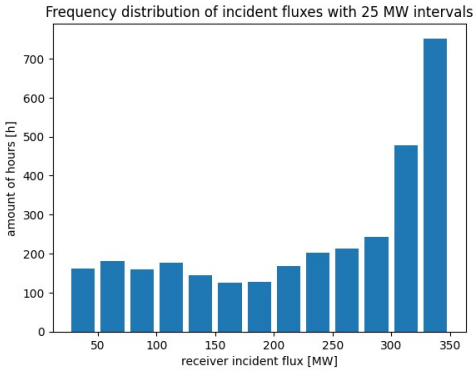
- η_{des} is the design point optical efficiency, equal to 0.7002.
- $f_{el}(\alpha_s)$ is a 6th-degree polynomial accounting for the elevation effects.
- $f_{az}(\gamma_s)$ is a 4th-degree polynomial accounting for the azimuth effects.

The complex interaction between these variables is visualized in the 3D Optical Efficiency Surface, Figure 3a. This surface plot explicitly demonstrates how efficiency drops significantly at low solar altitudes, primarily due to the cosine effect and varies symmetrically with the azimuth, reflecting the field's geometry.

Finally, the analysis evaluated the yearly availability of solar power at the receiver aperture. Figure 3b illustrates the yearly frequency distribution of incident thermal power, aggregated in 25 MW intervals. This operational profile drives the dispatch strategy and TES management, serving as the primary input for the thermal model detailed in the following chapter.



(a) 3D Optical Efficiency Surface generated via polynomial regression



(b) Frequency distribution of receiver incident flux throughout the year

Figure 3: Optical efficiency surface and annual receiver incident flux distribution

4 Thermal Model

4.1 Input data

From SolarPilot's simulations, we could characterize the thermal flux hitting the receiver. Since our receiver is designed at summer solstice at summer noon at $950W/m^2$, this is the peak flux that can happen in a year. After surface discretization, for simulations in offdesign (part-load simulations) we decreased the flux in each point by a factor K:

$$K_{flux} = \frac{Qrec}{Qrec, design} \tag{4}$$

Where $Qrec, design$ equals to 347173kW (peak flux is $949kW/m^2$ which is under the limit of $1MW/m^2$).

All geometries are completely defined [9].

4.2 Modelling

The receiver has a height of 15 m and diameter of 17.65 m , we then made chose to use 48 panels of 22 tubular receivers each, for a total of 1056 tubes where solar salts need to flow, heating up from 255°C (cold TES temperature) to 565°C, max operating temperature. Being that the chosen fluid melting point was at 220°C we took 35 degrees as a safe margin, but this margin could also have been greater [11].

We neglected cold fluid piping losses in the upcamer hydronic circuit, which we thought was an acceptable hypothesis since also for the hot salts thermal losses are not very important (and for a colder fluid these losses would also be even less). This helped us to have a constant inlet solar salts temperature.

Therefore we have an unknown mass entering at 255°C and exiting at 565°C. To solve the thermal model we have to solve two iterative loops:

1. external loop for the mass flowrate of salts: after a first hypothesis of mass flowrate we enter the internal loop, and check the outlet temperature: if the outlet temperature is 565°C, we went on to the next tube, otherwise we updated the mass guess.

2. internal loop for the temperatures: solving an energy balance for each small volume we could find the amount of power that was actually absorbed by the HTF.

At the end we would sum up all the mass flowrate of the tubes.

4.3 Thermal equivalent circuit

Thermal resistances for heat transfer included:

1. Internal convection,we used Gnielinski correlation with f being the friction factor:

$$R_{conv} = \frac{1}{h_{conv} A_{int}} \tag{5}$$

$$Nu = \frac{\frac{f}{8}\,(Re - 1000)\,Pr}{1 + 12.7\left(\frac{f}{8}\right)^{1/2}\left(Pr^{2/3} - 1\right)} \tag{6}$$

$$f = (1.82\log_{10}(Re) - 1.64)^{-2} \tag{7}$$

2. Heat conduction in a cylindrical wall:

$$R_{cond} = \frac{\ln(r_2/r_1)}{\pi dzk} \tag{8}$$

   For this resistance we thought it would be more appropriate to only use half of the surface, since only half of the tube external surface is exposed to the radiation, and the back side of the tube is simplified as adiabatic (that is why a 2 doesn't appear at denominator).

3. External natural convection:

$$Ra_L = \frac{g\beta(T_s - T_\infty)dz^3}{\nu\alpha} \tag{9}$$

   Then if $10^4 < Ra < 10^9$:

$$Nu_L = 0.59\,Ra_L^{1/4} \tag{10}$$

   else if $10^9 < Ra < 10^{13}$:

$$Nu_L = 0.10\,Ra_L^{1/3} \tag{11}$$

4. Radiative exchange:

$$Qrad = \sigma\epsilon A_{ext}(T_{wall}^4 - T_{sky}^4) \tag{12}$$

   With $T_{sky} = 0.0552 \cdot T_{amb}^{1.5}$

All non-dimensional numbers like Re,Ra,Pr where correctly calculated with their definition. After a first guess of absorbed heat, and therefore fluid mean temperature, we could calculate the wall temperature[2]:

$$T_{wall} = T_{mean}^{htf} + Q_{htf}(R_{conv} + R_{cond}) \tag{13}$$

From there losses are computed as :

$$Q_{losses} = h_{conv}^{ext} A_{ext}(T_{wall} - T_{amb}) + \sigma\epsilon A_{ext}(T_{wall}^4 - T_{sky}^4) \tag{14}$$

From there we needed to check the energy balance on the volume:

$$|\alpha Q_{rec} - Q_{htf} - Q_{losses}| < toll \tag{15}$$

With

$$\alpha = \frac{\alpha_{coating}}{F_{tt}(\alpha_{coating} - 1) - 1} \tag{16}$$

Where the absorbance takes into account for the view factor between adjacent tubes[2].

If the check was valid we went on to the next volume,otherwise we would update our Q_{htf}[2].

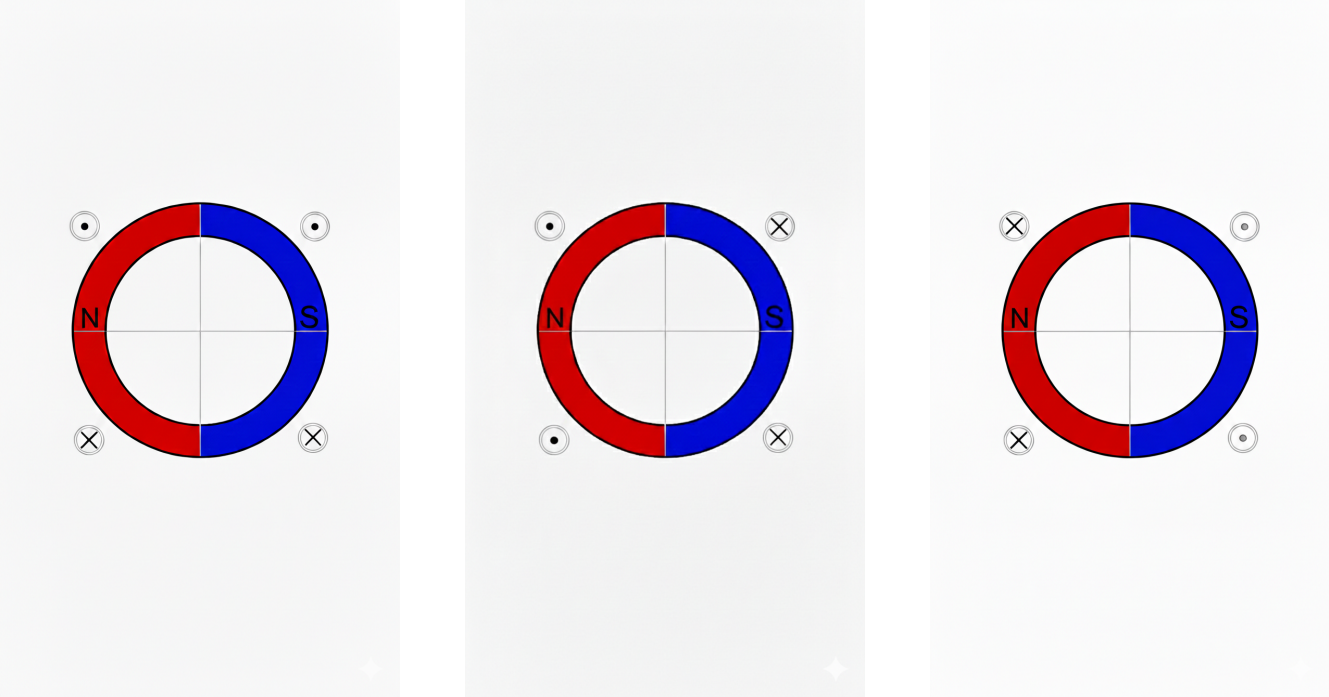
4.4 Results

Thermal efficiency is defined as:

$$\eta_{th} = \frac{Q_{htf}}{Q_{rec}} = \frac{\alpha Q_{rec} - Q_{losses}}{Q_{rec}} = \alpha - \frac{Q_{losses}}{Q_{rec}} \tag{17}$$

We run different simulations for three different configurations of the receiver: since solar salts flow upward and downward, we needed to find the optimal flow configuration[9]. Ideally dividing the receiver into one hot zone and one cold zone we define three alternative configurations:

1. Version 1: fluid flows upward and downward equally splitting hot and cold sides
2. Version 2: fluid flows upward in hot side and downward in cold side
3. Version 3: fluid flows upward in cold side and downward in hot side



(a) version 1 (b) version 2 (c) version 3

Figure 4: different version of reciever

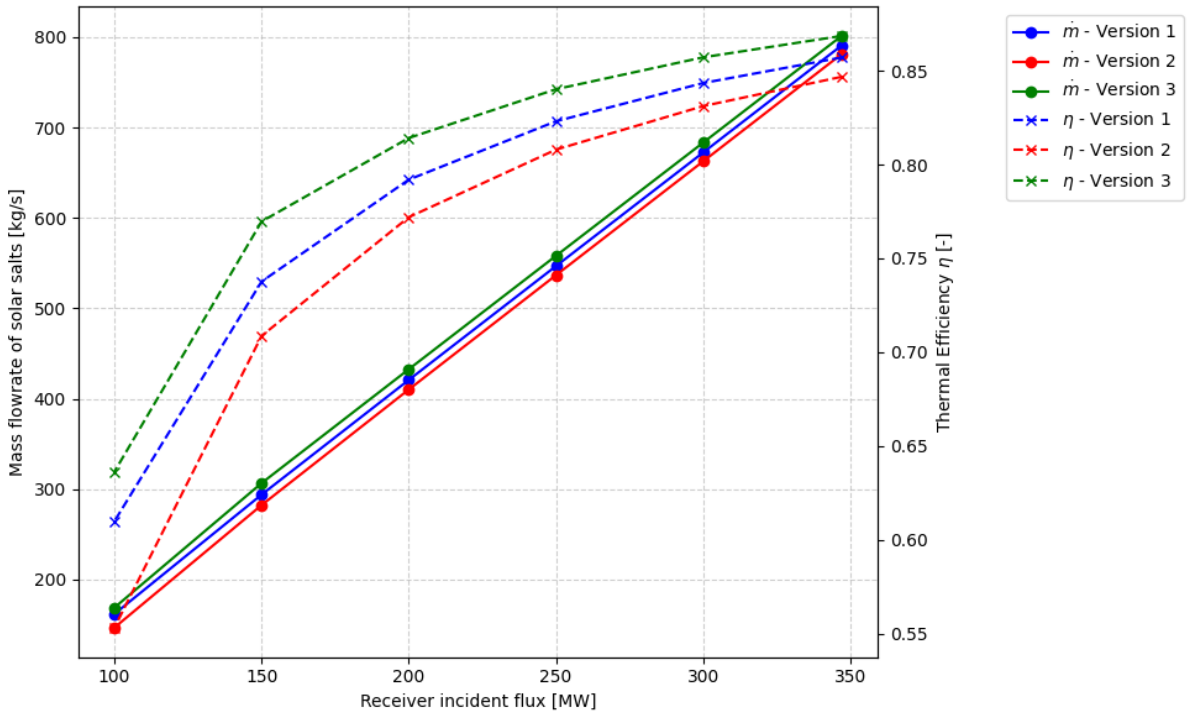


Figure 5: thermal efficiency and solar salts mass flow rates

In figure 5 we can clearly see how the third version of the receiver behaves better as it yields the higher flowrate at each receiver incident flux and the better thermal efficiency.

This is due to the fact that since the fluid flows downward in the hot side, it has much less "time" to lose its thermal energy; when the opposite happens (fluid flows upward in the hot zone) we have a hot fluid (high

temperatures of walls) losing energy for a bigger portion of total surface.

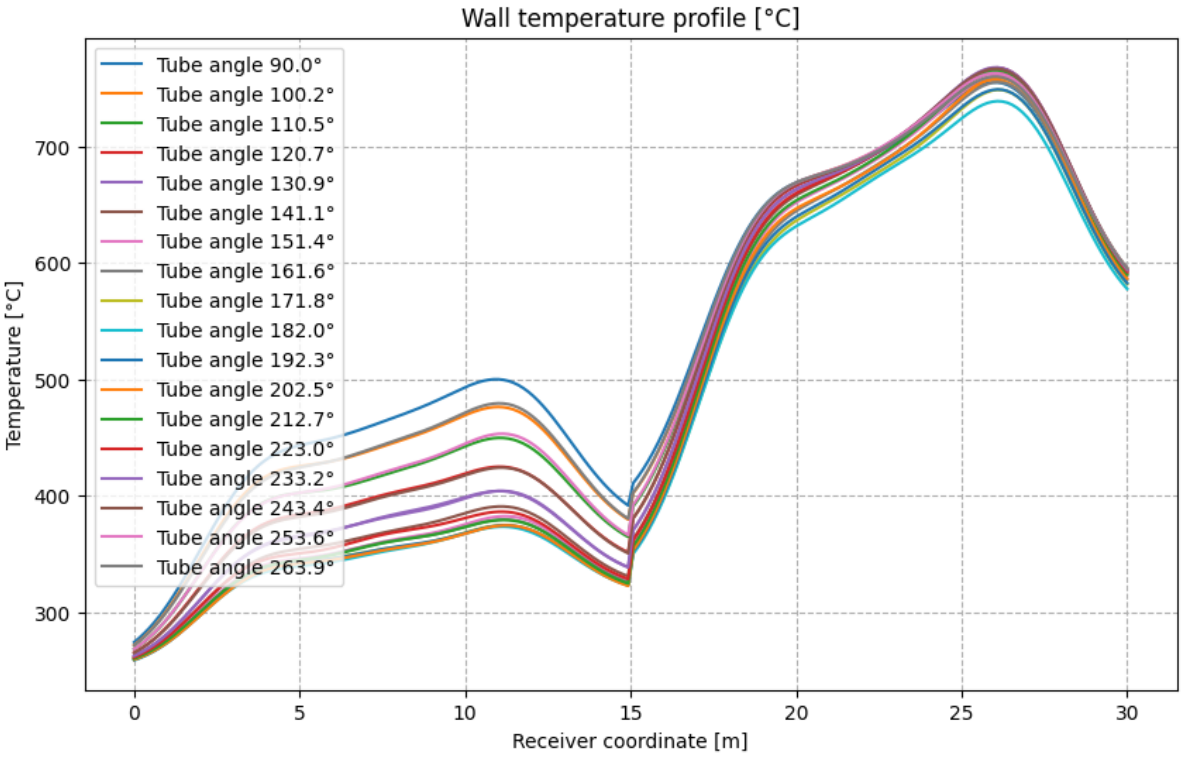


Figure 6: Wall temperatures for different tubes

In figure 6 we can clearly see how wall temperature changes for different tubes, it can be clearly seen how the fluid flows in the cold side first and then hot side as wall temperatures are much higher in the last 15 meters of flow (downward portion).

The temperature discontinuity in the middle is given by the fact that obviously surfaces between the first portion and the second portion are not directly connected , and the energy balance converges towards a different value due to a different (non-continuos) thermal radiation value.

In the first section, the hotter tubes are the ones at angles of 90° (with respect to the north which are the ones subjected to a higher radiation, meanwhile the colder ones are those around 180°, subjected to a very low radiation. The colder tubes also carry the bigger mass flowrate since in the hot zone they are the ones hit by the bigger radiation.

In the last part temperatures drop since radiation is significantly lower with respect to the center of the receiver so wall temperatures tend to decrease even if the fluid is still heating up.

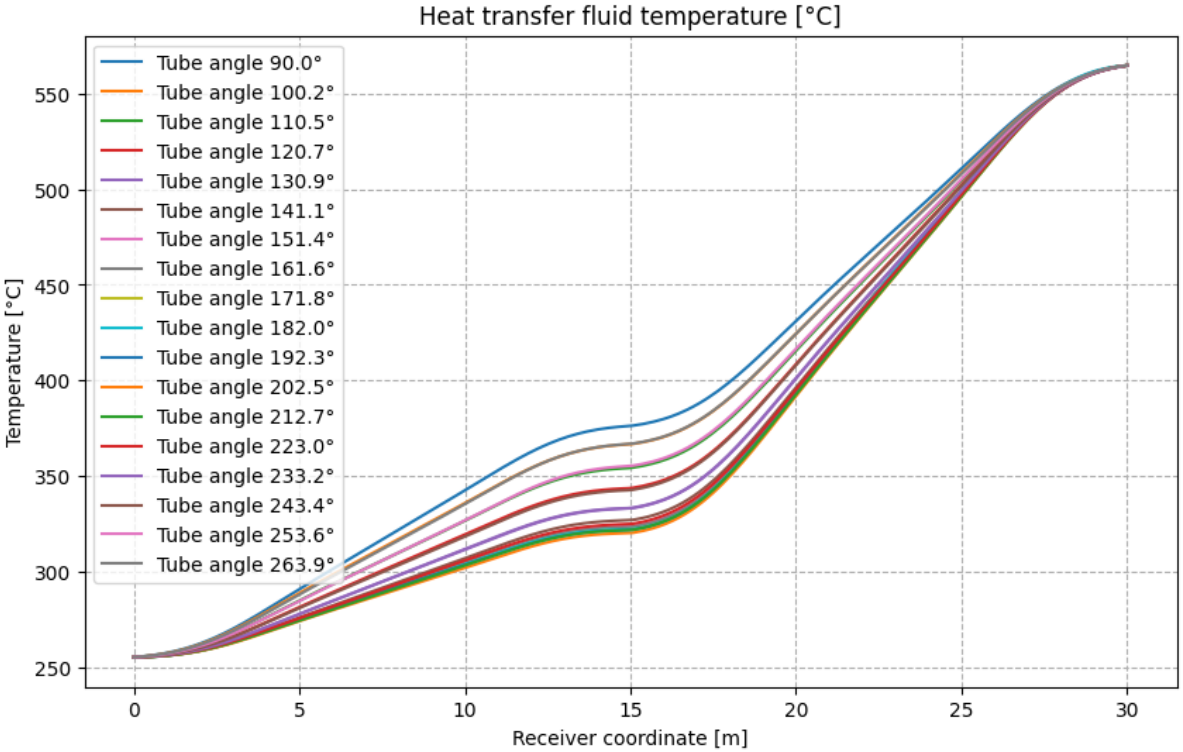


Figure 7: Heat transfer fluid temperatures

In figure 7 we see the heat transfer fluid behaviour with respect to its own temperature. We can note that the derivative of this plot is actually the thermal flux absorbed by the fluid; in the first zone we see higher flux in tubes at 90 degrees from north (the north side is the hotter one) and lower temperatures in tubes at 180 degrees from north. In the second side (downward flow) the flux is much higher for the tubes which were previously colder, meanwhile the smaller slope is given by the tubes that were the higher sloped in the first 15 meters of flow.

Convergency to 565°C is obtained for all tubes.

5 Power Block

The power block modeled is a Rankine superheated steam cycle that operates under the molten salts heat source, available between 565°C and 255°C. The computational model comprehends 8 heat exchanger(1 deaerator)for the feedwater preheating line followed by economizer, evaporator, and superheater. The expansion is made by the steam turbine and a air cooled condenser and a feedwater pump close the cycle.

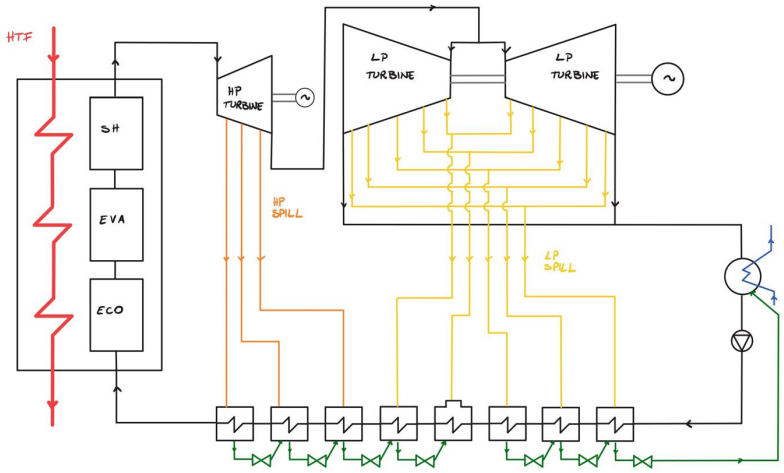


Figure 8: power block plant scheme

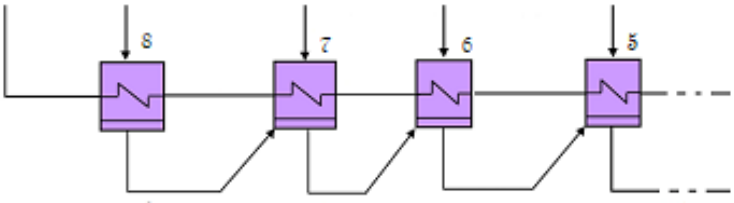
To achieve the higher electrical efficiency a iterative optimization of Peva (pressure of evaporation) has been made. Starting from the thermal input of the salt (mass flow rate and temperature range fixed),considering constant ambient conditions (Tamb=25°C) and defining pinch point, approach point and subcooling for each evaporation pressure the mass of steam generated and a residual quantity of heat are calculated (Qleft).

This quantity represents the thermal duty of the preheating line, and consequently deltaT, mass flow rate, temperature and pressure of steam spilled for each one of the 8 exchanger are automatically calculated. Laslty the expansion is performed considering all the bleedings and condenser and pump duty are realated to the mass of steam. The optimization is made considering all the mechanical,thermal,electric and fluid-dynamic losses and operating constrain(ex: expansion not permitted with too low quality of vapor)are considered. Result show an optimal value for evaporation pressure at 3.97 MPa and a related efficiency of 36.67% .

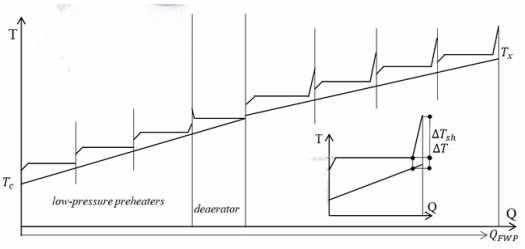
5.1 Preheating line and bleeded expansion structure

Starting from the whole water mass flow rate and the whole thermal duty,for each of the exchanger we defined a deltaT (1/8 of the overall preheating thermal increase) and a constant pinch point that defines for each exchanger pressure (and the temperature) of saturation of the steam extracted to target the bleeding point along the expansion (starting from the eighth with outlet temperature known) . For this pressure value, in the expansion there is steam available at a certain temperature value (superheated) and this set consequently the mass spilled (higher the temperature of the spilled steam lower the mass spilled due to heat balance). From the first to the penultimate exchanger,excluding deareator,(starting from condenser side) also the reinjection of liquid flow rate from the subsequent exchanger is considered.

Finally the entire bleeded mass in summed and subtracted from the expansion work in the turbine.



(a) preheating line structure



(b) preheating line T-Q diagram

6 Piping Losses

A steady-state thermal model has been made to estimate the temperature drop of the Heat Transfer Fluid (HTF) as it travels through the solar tower piping. The simulation is made considering a single equivalent pipe representing all the fluid path present in the reciever (methodology used in SAM). It accounts for a 195m vertical tube and a 15m horizontal. The eq. pipe's internal diameter (D_{int}) is fixed at the design point by maintaining a constant flow velocity ($v = 1.5$ m/s) relative to the design mass flow. The steel pipe wall thickness is defined as 4.5% of the external diameter to ensure structural integrity.

For insulation a 90mm layer of high-performance insulation ($k = 0.078$ W/mK) is applied to minimize thermal dissipation [1].

The heat transfer from the hot fluid to the environment is modeled using a discretized radial resistance network. The pipe is divided into 100 segments (dz), and for each segment, three type of thermal resistances are calculated. Internal convection resistance(R_{conv_int}) as before (thermal model)is calculated using the Gnielinski correlation for the Nusselt number, also like the internal wall conduction resistance (R_{cond}) of the steel($k = 16.3$ W/mK). Insulation conduction resistance is added(R_{is}), modeled using the logarithmic thermal resistance formula for cylindrical geometries.

The external boundary conditions (natural convection and radiation) depend on the unknown outer wall temperature (T_{wall}), our model employs an iterative solver for each longitudinal segment. The solver balances the internal heat flux against the external losses of natural convection(relatve numerical funcition)and radiative exchange (Stefan-Boltzmann law) considering the emissivity (ϵ) of the external jacket.

By integrating these losses across the 100 discrete elements, the model predicts the final temperature of the HTF (T_{out}) reaching the power block(in design conditions)

When off-design conditions occur, the mass flow rate of the heat transfer fluid decreases.

$$\dot{m} = \rho\, A\, v \tag{18}$$

Since the virtual equivalent geometry remains constant, the velocity decreases proportionally. This variation is physically comparable to an increase in the residence time of the fluid within the pipe, which leads to higher thermal piping losses and decreased thermal efficiency.

$$\eta_{th_{pip}} = \frac{Q_{htfout}}{Q_{htfin}} = 1 - \frac{Q_{piplosses}}{Q_{htfin}} = \frac{Tout - Tmin}{Tmax - Tmin} \tag{19}$$

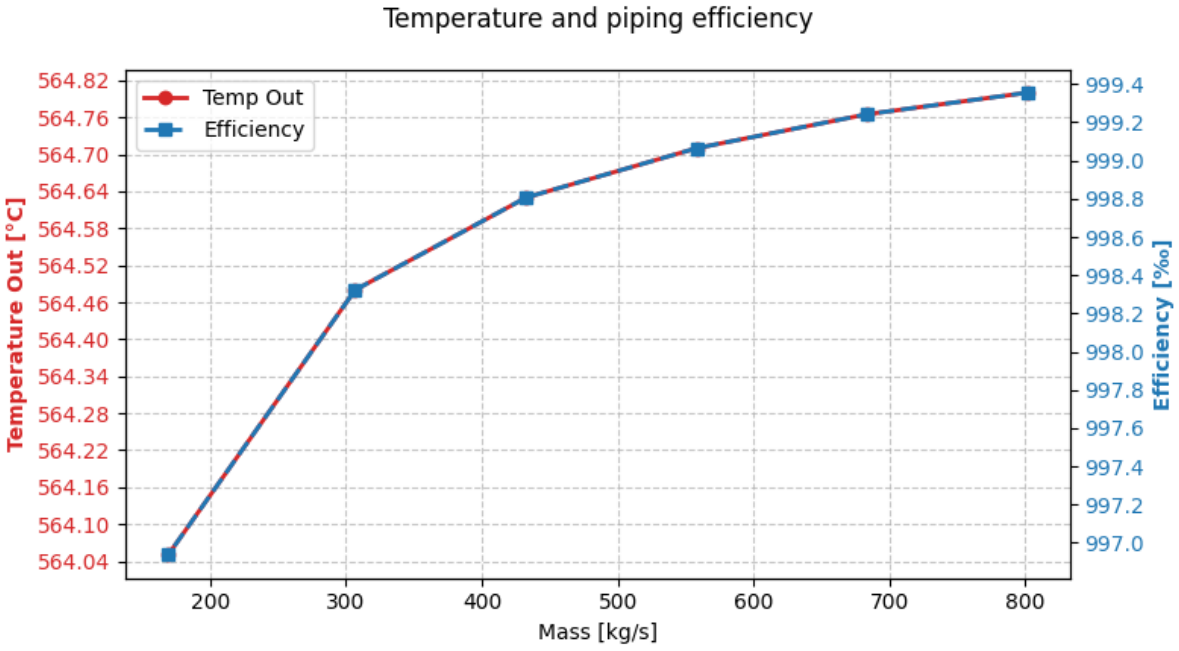


Figure 10: piping thermal efficiency at partial load

As the plot shows efficiency is only a function of outlet temperature, which is by itself only a function of mass flowrate, since in the piping efficiency the mass cancels out.

7 Thermal energy storage

The storage unit is modeled as a two-tank system (hot and cold) [3], where the total capacity is dimensioned in function of the cycle's nominal thermal power ($Q_{in,cycle}$) and the required hours of autonomy. The total volume of the single tank (V_{tes}) is calculated using the design mass flow rate and the required storage duration:

$$V_{TES} = \frac{\dot{m}_{design} \cdot 3600 \cdot h_{storage}}{\rho_{fluid}} \tag{20}$$



Figure 11: Tes structure

The tank dimensions are defined through an Aspect Ratio (AR), relating the tank height to its diameter, which is a critical parameter for determining the surface area available for heat dissipation. The model tracks the fluid mass during charging and discharging phases. A saturation logic is implemented to ensure that the stored mass (m_{stored}) never exceeds the physical capacity of the tank.The model treats each tank as a Continuously Stirred Tank Reactor (CSTR), assuming instantaneous mixing (lumped-parameter model).The temperature of the stored fluid (T_{tes}) is updated at each time step using a weighted energy balance between the incoming mass flow and the existing volume:

$$T_{tes,updated} = \frac{(m_{in} \cdot T_{in}) + (m_{stored} \cdot T_{tes})}{m_{in} + m_{stored}} \tag{21}$$

This approach ensures the conservation of energy during the fluid transfer processes between the solar field, the storage, and the power block.

7.1 TES thermal losses

To accurately predict the self-discharge rate, a thermal resistance network is established for the tank's boundaries[18]. The model accounts for a 30mm structural steel wall and a significant 400mm layer of mineral wool insulation . Heat loss through the lateral surface is calculated using the logarithmic thermal resistance formula for cylindrical layers. Conductive losses through the roof (top) and the floor (bottom) are modeled as plane-wall resistances [19] [20]. The model incorporates heat dissipation to the environment via natural convection(modeled with the previous used function) and accounting for the surface emissivity of the tank's external cladding. By integrating these losses, calculated using specific computational function taken from reference([19]), the model provides the final state of charge and the actual temperature available for the power cycle after storage for each hours of the year.

$$SOC(t) = \frac{m_{tes}(t)}{M_{max}} = \frac{m_{tes}(t)}{V_{tes} \cdot \rho_{htf}} \tag{22}$$

7.2 TES operating conditions

TES can operate in charge or discharge, depending by the fluxes of energy between the Solar Field and the Power Block.for each hour the priority is the continuous power generation achieved as far as possible through the following operating conditions:

1. **Excess Generation Mode**: When the hourly mass flow from the solar field exceeds the design requirements of the PB, the system operates at nominal capacity and the surplus thermal energy (htf mass flow rate in excess) is diverted to the hot TES tank.

$$m_{outSF} > m_{des-cycle} \quad \rightarrow SOC \nearrow \quad \rightarrow P_{el} = P_{des} \tag{23}$$

2. **Storage-Assisted Generation**: In cases of sub-optimal solar irradiance, the system attempts to bridge the gap by discharging the hot TES tank. If the accumulated mass is sufficient to meet the deficit, the power block maintains its nominal electrical output.

$$m_{outSF} < m_{des-cycle} \wedge m_{teshot} > (m_{descycle} - m_{outSF}) \quad \rightarrow SOC \searrow \quad \rightarrow P_{el} = P_{des} \tag{24}$$

3. **System Shutdown and Partial Charging**: If the combined energy from the solar field and the TES is insufficient to meet the power cycle's operational threshold, electrical production is set to zero. Any available solar-derived thermal energy is utilized to increment the storage level without engaging the power block.

$$m_{outSF} < m_{des-cycle} \wedge m_{teshot} < (m_{descycle} - massoutSF) \quad \rightarrow SOC \nearrow \quad \rightarrow P_{el} = 0 \tag{25}$$

4. **Defocusing**: When TES is completely charged and cannot absorbes more energy, if solar field exceeds the designpower, defocusing is adopted dissipating the surplus of energy.

$$m_{outSF} > m_{des-cycle} \wedge SOC = 100\% \quad \rightarrow P_{el} = P_{des} \quad \rightarrow defocusing \tag{26}$$

8 Auxiliaries consumption

In this section we take into account exclusively for the solar salts pump, as the feedwater pumps consumption is included in the power block efficiency (and is anyway of an entire magnitude smaller than the solar salts pump consumption).

Being that the TES tanks operate at atmospheric pressure, the solar salts pump has to win the entire geodetic prevalence given by the tower height (195m).

$$\Delta H = H_{height} + H_{friction} \tag{27}$$

Since we have 180m tower with a 15m receiver at the top we can split the friction prevalence considering an upcamer tube carrying all the solar salts mass, with 180m height (plus a 20m horizontal tube from the cold TES) and the friction losses for the 15 meters of receiver tubes where accounted for by taking into account for the losses of the worst operating receiver tube, as far as friction losses go. We used the same friction factor employed in the thermal model, the Petukhov correlation, which is valid for turbulent flows.

$$f = (1.82 \log_{10}(Re) - 1.64)^{-2} \tag{28}$$

$$\Delta p_{pump_friction} = f \cdot \frac{height}{D_{int}} \cdot \frac{\rho \cdot v^2}{2} \tag{29}$$

Since space velocity of the fluid was set in the downcamer tubes to 1.5 m/s in design condition (so in the offdesign condition this velocity would be lower); we kept the same design condition for the upcamer tube, therefore friction losses at such low velocities are not really important(only a couple of meters of prevalence).

For the last 15 m of the receiver in height we need to take into account for the head losses given by friction in the tube with the highest flowrate. This tube turned out to be the tubes which is invested by the highest flux in the downcamer part of the receiver, so the tube at 180° in the upcamer part (see figure 7). Results showed that the friction given by the worst operating tube could be entirely neglected as in the design point, with worst operating condition, the tube with the highest friction losses needed 3.7 cm of head.

Since its calculation in the offdesign condition only needed to solve the thermal model for 1 tube in part-load (to find the mass flowrate) we anyway decided to take it into account, even though the friction head loss was about 3.7 cm in design condition (very much negligible with respect to the geodetic head of 195m).

As can be seen in the figure 12 , which represents the power required by the pump of solar salts, the power is very high , even when the flowrate is low, actually the flowrate has very little effect in magnitude as the height of the tower is the major component of the energy needed.

That curve also represents the required head of the pumping system. Therefore we now have an auxiliary consumption function of only the mass flowrate of solar salts (that is defined by solving the thermal model).

$$P_{aux} = \rho \cdot g \cdot \Delta H_R / \eta_{hydr} \tag{30}$$

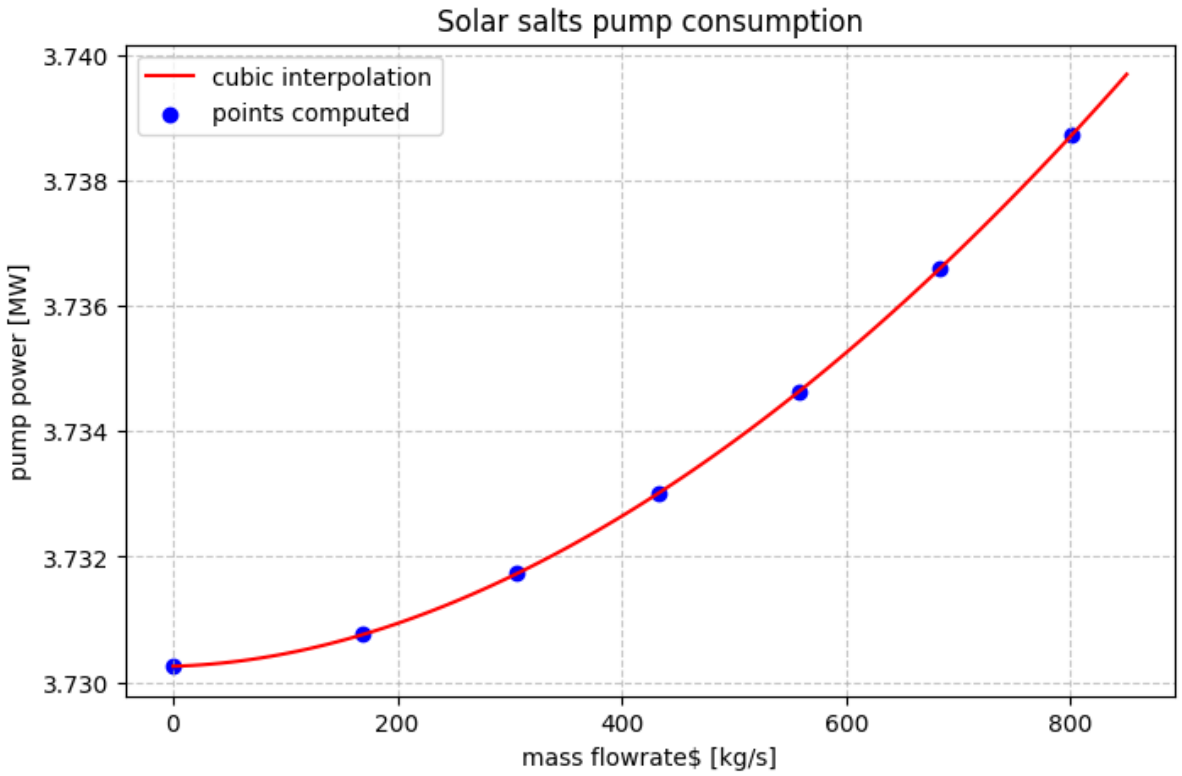


Figure 12: Pump power

During simulations, based on the height os solar salts in the cold tank, we can subtract required head from the hydraulic circuit, so in the simulations auxiliaries power was computed as :

$$P_{pump} = P_{aux}(\dot{m}_{salts}) - rho \cdot g \cdot h_{salts}^{coldtank} \tag{31}$$

The height of the salts in the cold tank basically gives free pumping power.

Height of the colomn was computed in a clever way with the cold SOC.

$$h_{salts}^{coldtank} = SOC^{cold} \cdot H_{TES} \tag{32}$$

9 Optimization

Optimization simulation was done to minimize LCOE (levelized cost of electricity),defined as

$$LCOE\,[\frac{\$}{MWh}] = \frac{CAPEX * CRF + fixedOM}{AEP} + varOM \tag{33}$$

Capital recovery factor is defined as

$$CRF = \frac{i \cdot (1+i)^n}{(1+i)^n - 1} \tag{34}$$

With lifetime (n) equal to 25 years and inflation (i) set to 8%.[5] For CAPEX costs, following data was taken:

For OPEX costs, following data was take:

Economic data was taken from a vast pool including [5],[6],[13],[17],[?],[21].

Simulations were carried out at variable SM and TES hours, with SM varying from 1 to 5 with 0.2 increment (later modified from 1.6 to 5 , same increment) and hours of storage varying from 5 to 18 hours of storage (later changed to 7 - 16 hours).

The simulation was carried out hourly for an entire year and followed these steps:

1. From DNI,ambient temperature

$$Q_{sun} = DNI \cdot A_{SF} \tag{35}$$

Parameter	Value
Power block	900 $[\frac{\$}{kW_{el}}]$
BOP and eng.	600 $[\frac{\$}{kW_{el}}]$
SF,receiver,tower	225 $[\frac{\$}{m^2}]$
TES	40 $[\frac{\$}{kW_{th}}]$
contingencies	14% CAPEX $[\frac{\$}{y}]$

Table 1: CAPEX values

Parameter	Value
Fixed OM	66 $[\frac{\$}{kW_{el}}]$
Variable	21 $[\frac{\$}{MWh_{el}}]$

Table 2: OPEX values

2. From the optical correlations detailed in the Solar Field chapter we knew solar altitude and azimuth for each hour of the year, coherently with SolarPilot convention, therefore we knew optical efficiency and receiver incident thermal power

$$\eta_{opt}(\alpha_s, \gamma_s) = \eta_{des} \cdot f_{el}(\alpha_s) \cdot f_{az}(\gamma_s) \tag{36}$$

$$Q_{rec} = Q_{sun} \cdot \eta_{opt}(\alpha_s, \gamma_s) \tag{37}$$

3. Also from the thermal model we knew thermal efficiency and the mass of solar salts for each receiver incident flux (figure 5)

$$\eta_{th} = \tilde{\eta_{th}}(Qrec) = \tilde{\eta_{th}}(h) \tag{38}$$

$$Q_{htf} = Q_{rec} \cdot \eta_{th} \tag{39}$$

$$\dot{m}_{salts} = \tilde{\dot{m}_{salts}}(Qrec) = \tilde{\dot{m}_{salts}}(h) \tag{40}$$

4. Knowing the mass of solar salts, piping losses (and temperatures at the outlet of the solar field) are completely defined from the piping losses model

$$T_{outSF} = T_{outSF}(\tilde{\dot{m}_{salts}}) = \tilde{T_{outSF}}(h) \tag{41}$$

Which is also the temperature of the fluid entering either the power block or the hot storage.

It is fundamental to understand that all previous quantities are fixed once the previously discussed models are correctly run, therefore each of these quantities is fixed every hour of the year (the only quantities that can change are receiver thermal flux (and consequently htf thermal power) in the case of defocusing).

In case of defocusing we decided to not modify also Q_{sun} because it would be easier to distinguish between defocusing and DNI drop in the hourly plots (therefore purely for a graphic reason, but this will be discussed better later).

Knowing these quantities are fixed for each parametric simulation we can run, we decided to save these numbers, in order to greatly decrease computational time.

5. SM definition:

$$SM = \frac{Q_{htf}^{des}}{Q^{in,cycle}} \tag{42}$$

In the definition of solar multiple we used heat transfer fluid design thermal power, not thermal power after piping losses since the procedure is more straight-forward and piping losses are anyway defined for each hour of the year (ie:results of the simulation would not change).

Changing SM with a fixed solar field would mean changing thermal power to the power block, therefore power block size.

6. hours of storage definition:

$$E_{TES} = Q^{in,cycle} \cdot h_{storage} \tag{43}$$

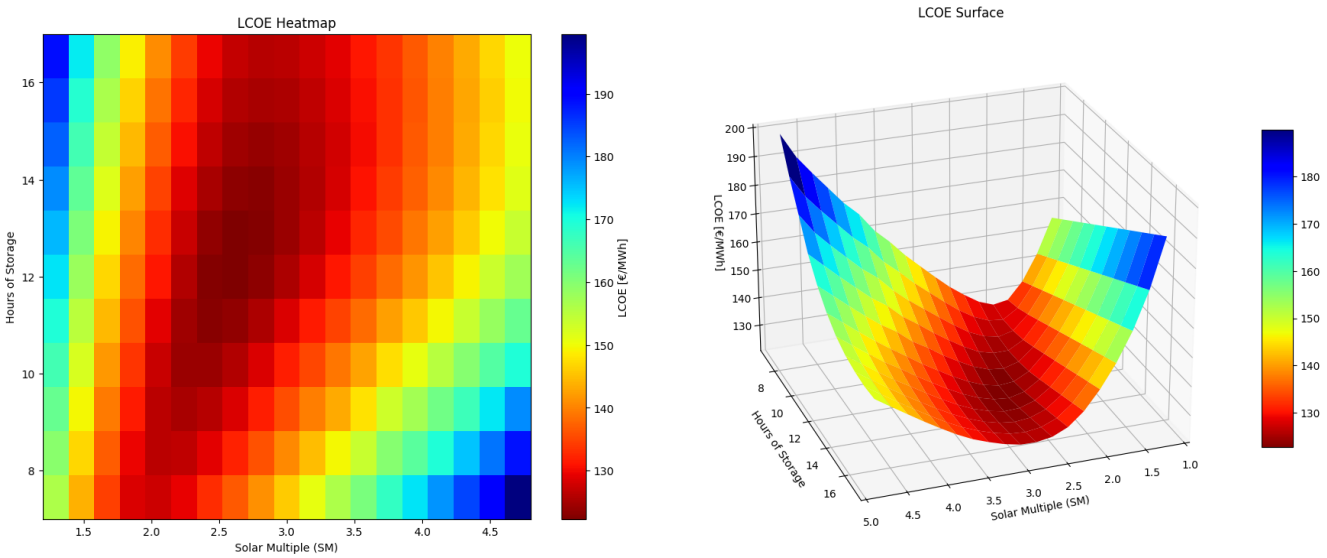
Since TES energy and therefore volume is function of both thermal input to the cycle and hours of storage , it is fundamental to run simulations at fixed SM for each hour of storage, then change SM in the outer "shell".

TES behavior during simulations was detailed in the previous chapter.

7. Auxiliaries consumption: With respect to auxiliaries consumption we found in the previous chapter salts pump consumption as merely function of mass flowrate of salts.

$$P_{aux} = \tilde{P_{aux}}(\dot{m}_{salts}) = \tilde{P_{aux}}(h) \tag{44}$$

9.1 Results



(a) LCOE heatmap

(b) LCOE surface

Figure 13: Results of LCOE optimization simulations

In figure 13 we can clearly see a minimum for the LCOE for solar multiples around 2.6 and for high hours of storage around 13 to 15 hours of storage.

A more dense parametric simulation highlighted an optimum for $SM = 2.6$ and hours of storage $= 13$.

For low SM and low hours of storage the investment cost for the "big" power block would be too high, and the AEP would be too low due to low size of storage.

Meanwhile for low SM and very high hours of storage the cost of the storage would be too high, and the AEP marginal benefit of having a larger TES decreases; once equivalent hours are really high we don't really need a larger TES, the cost-benefit tradeoff would be very negative. This would be overall the option with the highest costs (big PB, big TES) as can be also seen by the steep surface in figure 13b.

For high SM with small storage (small power block, small storage) TES operation would not fit well with the power block since as soon as DNI drops, the TES would quickly discharge, equivalent hours would be too low.

For high SM and big storage,again, the TES cost problem onsets once equivalent hours start to become very high.

9.2 Optimized configuration

The optimized configuration with SM= 2.6 and 13 hours of storage is has the following performance:

In the second colomn all relevant powers are computed as function of optimized SM and TES size, meanwhile in the third colomn we can see design point optical efficiency, thermal efficiency, piping efficiency, PB efficiency and auxiliaries efficiency.

Parameter	Value	Efficiency
Q_{sun}	495.82 [MW]	-
Q_{rec}	347.17 [MW]	0.7002
Q_{htf}	301.55 [MW]	0.8686
Q_{pip}	301.37 [MW]	0.9994
$Q_{in_cycle_des}$	115.98 [MW]	-
P_{gross}	42.53 [MW]	0.3667
P_{net}	38.79 [MW]	0.9121

$$\eta_{s-e} = \eta_{opt}\eta_{th}\eta_{pip}\eta_{PB}\eta_{aux} = 0.2032 \tag{45}$$

Solar to electric efficiency can be calculated consequently.

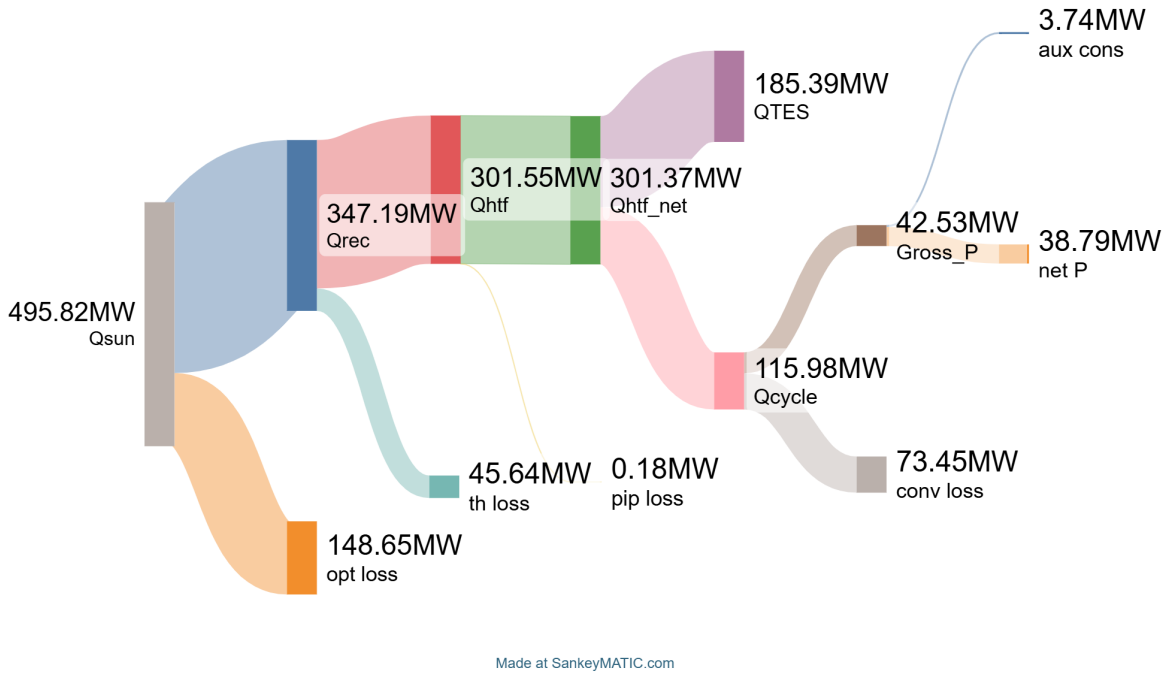


Figure 14: Power Sankey diagram

The only thing to be noted, besides the usual flow diagram from sun power to electric power is the split given by the solar multiple, which oversizes the solar field with respect to the power block by a factor of 2.6, sending thermal power to the thermal energy storage, to be used later when DNI drops.

9.3 Annual performance

Annual performance is summarized by the following table:

Yearly optical efficiency (= 0.5994) is heavily penalized by defocusing since (as already discussed) in our simulations defocusing affects receiver energy. Defocusing is heavily applied especially in the spring and summer seasons, as will be later shown by the hourly simulations plots.

Yearly thermal efficiency (= 0.8374) is high but as we can see in figure 3b the receiver operates around design point very frequently during the year, so we can expect this type of efficient operation for the receiver.

Particular attention has to be put into the power block efficiency, the yearly efficiency is higher than the design point ($\eta_{PB}^{des} = 0.3667$), this is due to the fact that we implemented a variable efficiency of the power block as function of the ambient temperature.

Table 3: CSP plant annual performance

description	**Value (GWh)**	**Yearly efficiency**
Sun Energy	1453.99	-
Receiver Energy	871.59	0.5994
Absorbed Energy	729.85	0.8374
Piping Energy	729.22	0.99914
Gross Energy	273.75	0.3759
Net Energy	241.68	0.8828
Thermal Energy Loss	0.90	0.9988
performance	**value**	
Equivalent Hours	5682.46 h	
LCOE	122.31 $[\$/MWh]$	

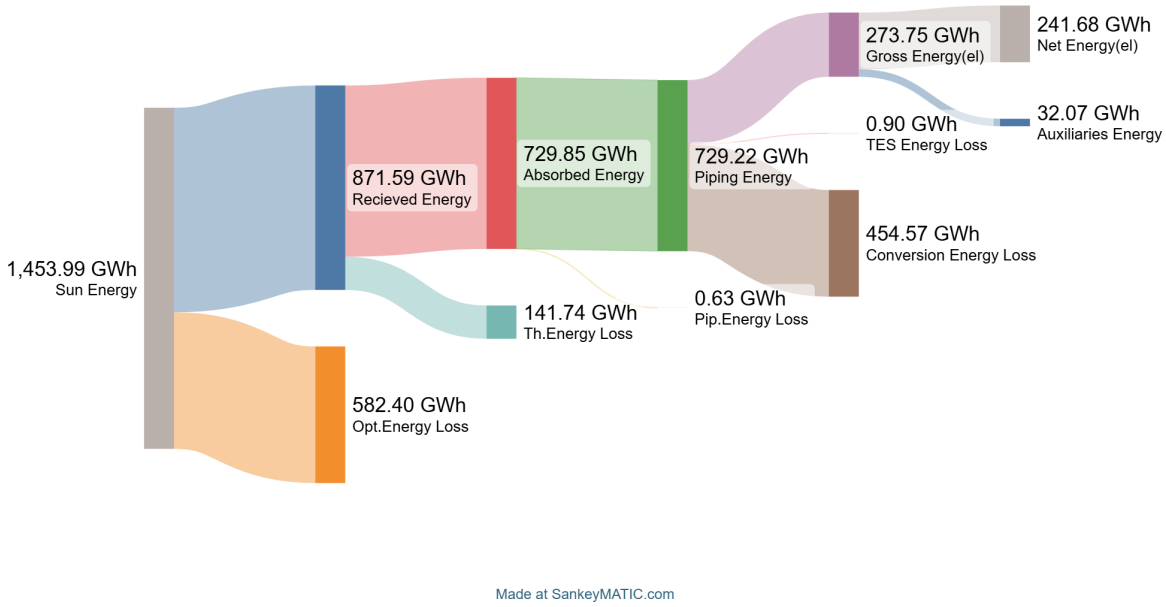


Figure 15: Annual energy Sankey diagram

This was carried out to take into account for the operation of the air-cooled condenser, being that its operation is heavily influenced by the ambient temperature.

Since we have an optimized, big thermal energy storage system, we can expect to run the power block in favorable conditions at night or during winter (design point was computed at 25°C), this can explain the improvement of the efficiency.

For the yearly operation we can also see how the thermal insulation efficiency appears:

$$\eta_{ins}^{th} = \frac{E_{in,cycle}^{yearly}}{E_{piping}} = 1 - \frac{E_{loss}^{th}}{E_{piping}} \tag{46}$$

This loss doesn't appear in the instantaneous power balance, as it is a loss given by the TES operation; the yearly energy losses appear with the yearly energy balance, as the energy losses affect the TES operation; with good insulation these losses can be minimized and we can provide a good thermal performance.

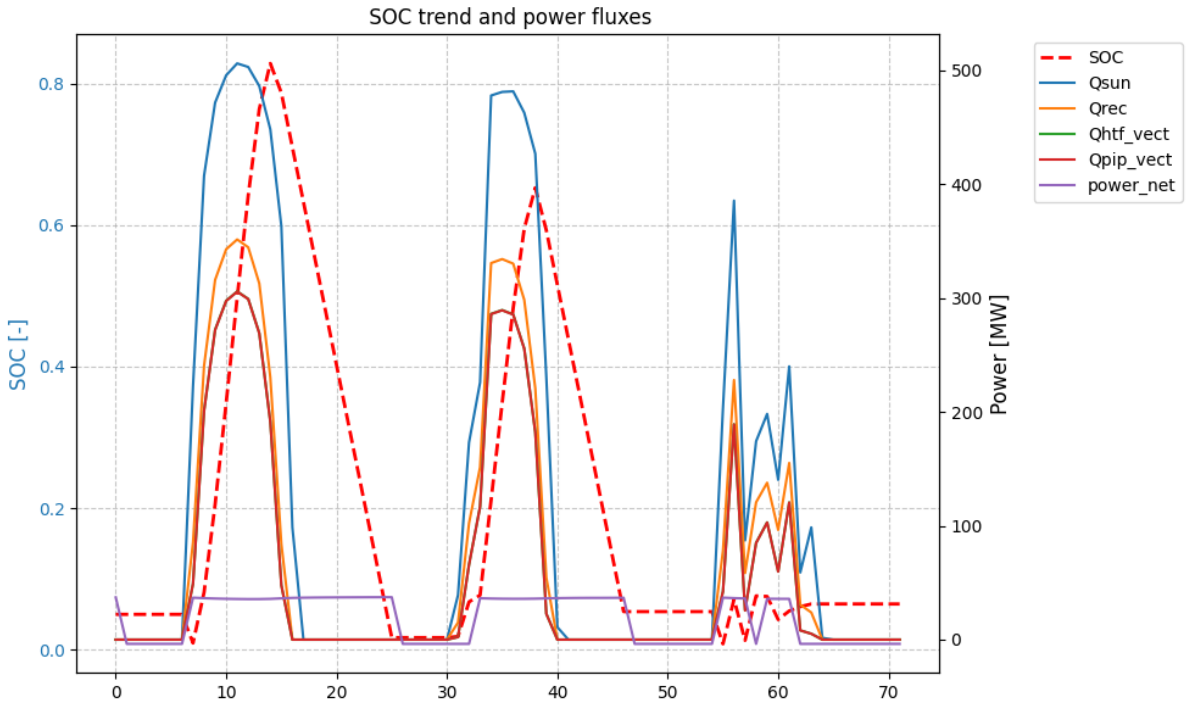


Figure 16: Winter day operation

9.4 Daily operation

9.4.1 Winter days

This figure characterizes operation between 5th - 8th of January. As we can see in figure 16 winter optimized operation still manages to run the power block for about 50% of the hours, and TES SOC (which is the only parameter to be read on the left vertical axis) is optimized to exploit low DNI in the winter condition to provide the power block with useful heat also after Sun irradiation stops.

Another peculiarity can be seen between the end of the second day where SOC is higher than zero (so hot TES is slightly charged) but not charged enough to run the power block for an entire hour (simulation is done on hourly basis).

On the third day bad weather caused irregular DNI and we finished the third day with discharged storage (not enough salts to run PB).

9.4.2 Spring days

In this plots we can clearly see the optimized operation of the plant, and the reason as to why yearly optical efficiency is not so high:defocusing.

In these days the power block os constantly run for 3 consecutive days (even at night with low ambient temperatures helping PB efficiency greatly).

When defocusing is activated receiver thermal power only need to cover for the thermal power needed by the steam cycle, as the TES is already fully charged. This type of optimized operation shows why increasing storage hours even more gives decreased marginal benefits, as the operation already employs defocusing in the spring season.

9.4.3 Summer days

This figure takes into account for days between 15th-18th of August. The first day starts with discharged TES and doesn't employ defocusing besides in 1 hours, and this first day charges storage to fully run an entire night, discharging fully just as DNI starts to ramp up in the second day.

The second day employs defocusing for a greater part of the operation, and this will come in hand in the third day where sun conditions are not satisfactory.

Power block successfully runs for three consecutive days, shutting down after a badly conditioned third day.

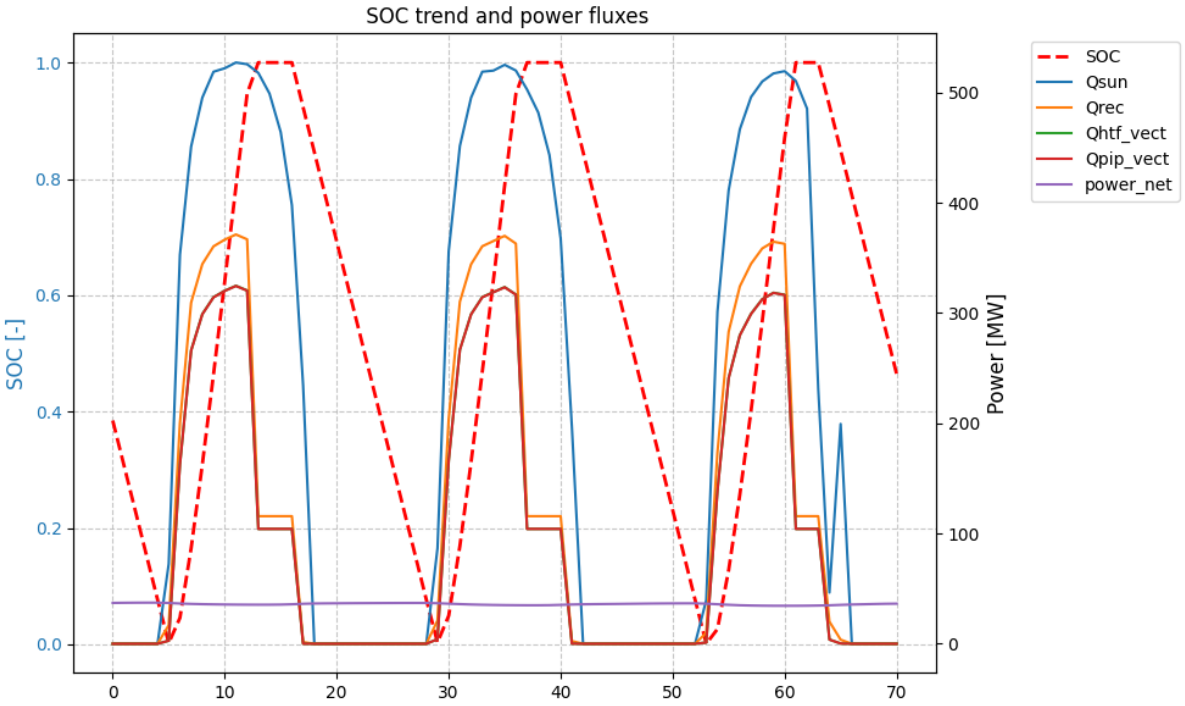


Figure 17: Spring day operation

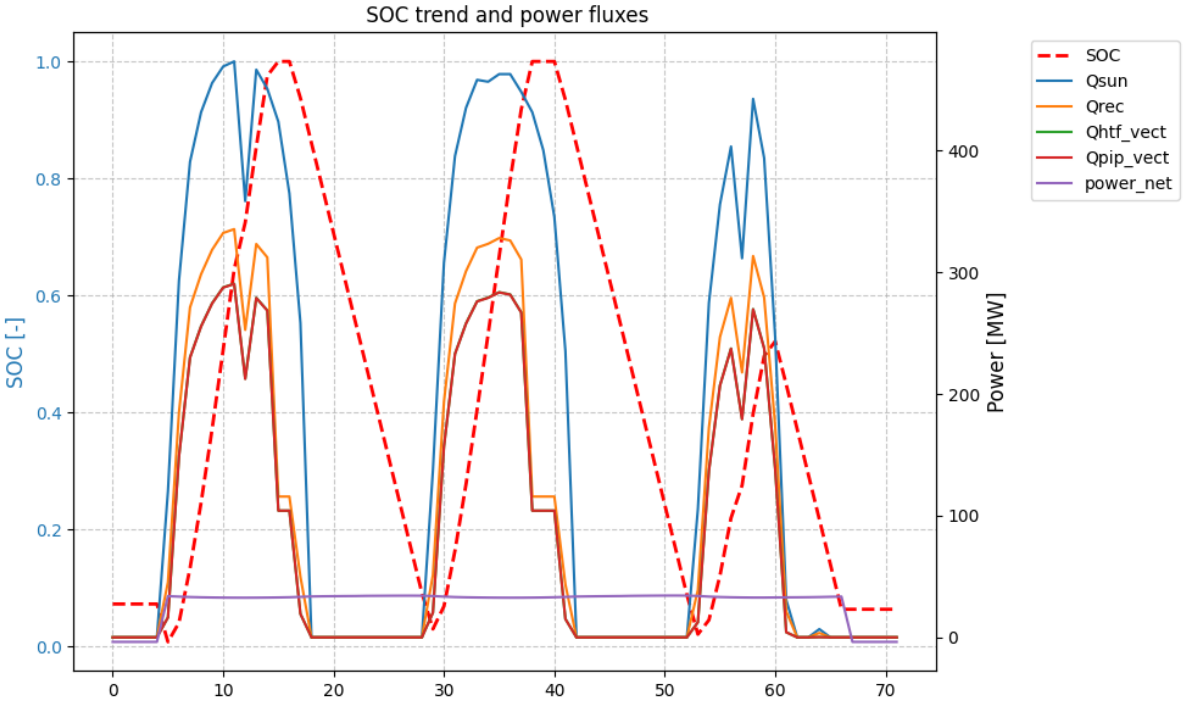


Figure 18: Summer day operation

9.4.4 Autumn days

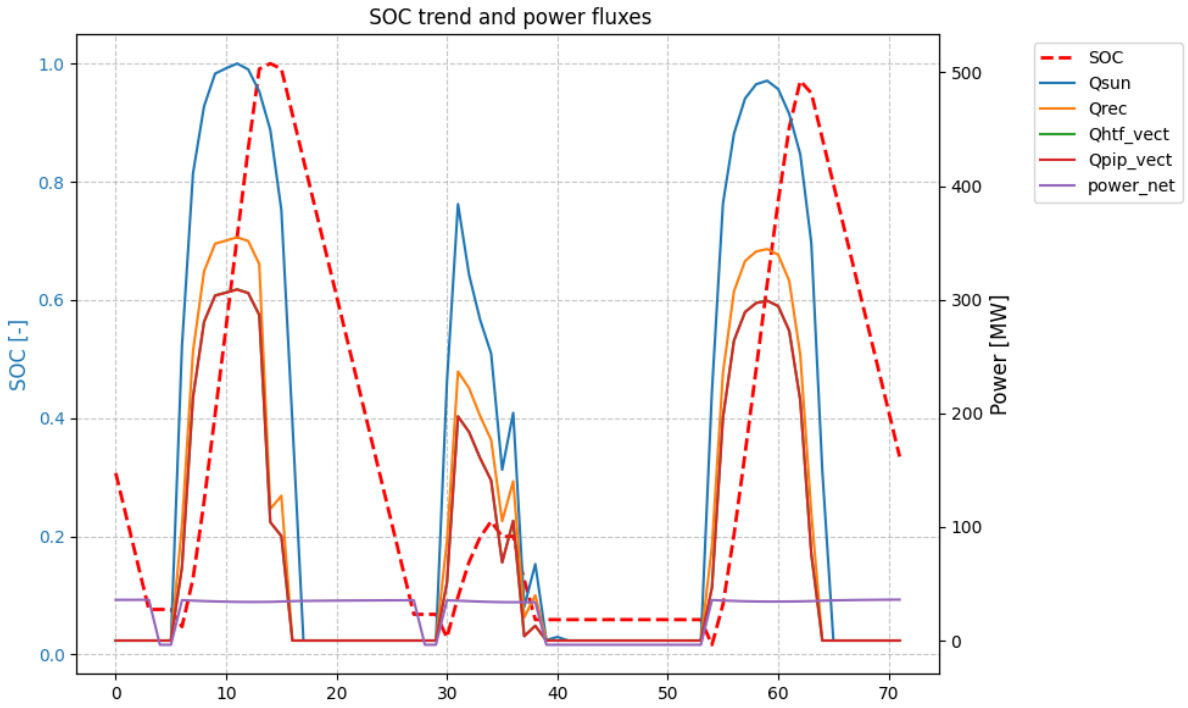


Figure 19: Autumn day operation

This plots shows operation between 23rd-25th of October, the 24th day shows bad conditions, but anyway the power block still manages to run fully for most of the time.

9.5 Cost breakdown

These pie charts represent the cost breakdown in detail for the LCOE separated by voices



(a) Percentages on LCOE

(b) Absolute portions of LCOE

Figure 20: Cost breakdown of the LCOE

All values on figure 20b are intended in $\$/MWh_{el}$ (software used did not let us display units of measure).

For the variable OPEX we found a source which gave us directly variable OPEX costs for a CSP plant in the United States so we didn't account for any other voice besides fixed $O\&M$.

All percentile voices are in line with literature data[13].

10 Conclusions

This project developed an integrated modeling framework for the design and optimization of a solar tower Concentrated Solar Power (CSP) plant, including solar field performance, receiver thermal behavior, piping losses, Thermal Energy Storage (TES), power block operation, and auxiliary consumption. The system was simulated on an hourly basis over a full year to evaluate both design and off-design performance.

The optimization identified an optimal configuration with a Solar Multiple of 2.6 and 13 hours of storage, representing the best compromise between investment cost and energy production. The optimized plant delivers 241.68 GWh/year, corresponding to about 5680 equivalent operating hours, with a solar-to-electric efficiency of approximately 20% and an LCOE of 122.31 $/MWh.

Results confirm the key role of thermal storage in ensuring dispatchable operation and maintaining power generation during low-irradiance periods and nighttime. While high solar availability leads to defocusing losses, the analysis shows that increasing storage beyond the optimal size would provide limited benefits compared to the additional cost.

Overall, the study demonstrates the importance of integrated techno-economic optimization to balance solar field size, storage capacity, and power block rating. The developed methodology provides a solid basis for further analyses under different locations, cost assumptions, or operational strategies.

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